

When does a deep-learning streamflow forecast beat the gauge?

Google’s open flood-forecasting LSTM on 28 California basins

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Abstract

River forecasts are often scored against a basin’s mean flow. An operator has a harder baseline to beat: the most recent gauge reading. On the 2017 and 2023 flood years, this weather-driven LSTM was worse than gauge persistence at one day, better on the cohort median at two days, and better on every basin from day four. Further tests cover the places where that advantage might fail: large storms, flood peaks, an unseen basin, predictive intervals, and NOAA’s process model. The model has useful two-to-seven-day skill, but only within clear limits.

Methods, configs, and reproduction notes: `docs/METHODS.md`.

The short version

On the evidence here, the LSTM is mainly useful as a two-to-seven-day volume forecast for gauged basins under conditions similar to those in training. It is not a reliable flood-peak predictor. At one-day lead it beats plain gauge persistence on 9 of 22 basins and damped persistence on 4 of 22; from day 4 it beats both on all 22. Its raw 90% intervals cover only 66–72% of observations, although a cross-fitted calibration raises coverage to 88–91% with almost no loss of sharpness. The advantage narrows on the largest storm days and appears between day 2 and day 4, depending on how the errors are counted. A basin excluded from training failed outright, for reasons this experiment cannot identify. Results against NOAA’s process model change with the model version and test years. The rest of the paper puts numbers on those limits.

The task, the model, and the score

The target is daily average streamflow at a basin outlet, from the current day through the following week. Each day ahead is a lead: lead 0 estimates today’s flow using today’s weather, while lead 3 is a three-day-ahead forecast. In California, the operational version of this problem is reservoir inflow—how much water will arrive as a storm approaches. The two-to-seven-day range studied here overlaps the decision window for forecast-informed reservoir operations (FIRO). The basins in this study are more varied than a reservoir-inflow cohort, as described below.

The model is an LSTM, a recurrent neural network that reads one day at a time and carries a learned state from one day to the next. Here, that state stands in for water stored in snowpack and soil. The inputs are basin-average precipitation and temperature from three weather products (ECMWF HRES forecasts, NASA’s IMERG satellite precipitation, and NOAA’s CPC gauge analysis), along with fixed catchment attributes such as area, mean precipitation,

aridity, snow fraction, and moisture and evaporative indices. The model never sees observed streamflow; it runs on weather and basin attributes alone. It predicts flow at leads 0 through 7 in mm/day, a water depth spread over the basin area so that large and small basins share a common scale. One output head gives a single value and trains on mean squared error. A second gives a probability distribution and trains on likelihood. Both use Google Research’s open flood-forecasting framework, the same model family used by Flood Hub (Nearing et al., 2024), although this 28-basin model is not the production system.

The model was trained on 28 minimally regulated California basins from Caravan (Kratzert et al., 2023), a community dataset of gauged catchments drawn here from the CAMELS subset (Newman et al., 2015). The lack of dams is deliberate: reservoir operators need forecasts of unimpaired inflow, and near-natural gauges measure a similar quantity, although only some of these basins drain toward managed reservoirs. The geography extends well beyond the Sierra. Under the snow-fraction rule used here (basins below 0.3 are classified as rain-dominated), the set contains 20 rain basins and 8 snow basins. Most of the rain basins are in North Coast systems such as the Eel, Smith, Trinity, Mad, and Redwood, along with six Central Coast streams. The snow basins are Sierra and Tahoe headwaters: the Merced pair, Bear Creek, Pitman Creek, the upper Trinity, and three Tahoe-area gauges. Caravan’s published streamflow records end in 2014 and were extended through October 2024 with USGS data.

There is an important complication in the weather inputs. HRES begins in mid-2012 and GraphCast in 2016. Before those archives begin, the pipeline substitutes ERA5-Land reanalysis shifted to the target date—realized weather in place of a forecast. Training from 1985 to 2016 therefore used nearly perfect future weather in those channels for most dates, whereas the flood test years use actual forecasts. The training data are easier than the test data. Comparisons across eras, including the drought window and the 1997 flood below, consequently mix hydrologic differences with differences in forecast realism.

Forecasts are scored by NSE, the Nash–Sutcliffe efficiency, hydrology’s standard skill score. A score of 1 is a perfect match to observations, 0 means no more accurate than always predicting the basin’s mean flow, and negative values are worse than that constant.

The test

The test years are water years 2017 and 2023. (A water year runs from October through September and takes the year in which it ends.) Both were atmospheric-river flood years. The WY2017 sequence damaged both spillways at Oroville Dam and forced the evacuation of nearly 190,000 people downstream. WY2023 brought the December–January storm train that ended a three-year drought. Training uses the rest of the record, apart from the 2009–2011 validation years, and checkpoint selection is based on validation skill rather than the final epoch. Twenty-two of the 28 basins have observations in the flood windows. Unless noted otherwise, the medians below use that fixed cohort. Finding 5 instead uses five focus basins: four snow-classified and one rain-classified.

This design has two qualifications. First, WY2017 and WY2023 are challenge years, not untouched tests. They were selected for their extremity, and their results later influenced model capacity, the probabilistic analysis, and the framing of the project. Second, the train/test split is contiguous in forecast issue dates despite the seven-day horizon. At the split boundaries, 28 dates—all in the first week of October—appear as labels in both a training sample and a test sample. They account for 1.9% of scored rows and fall in reliably low-flow periods, so their effect

The 28 training basins: 20 rain-classified, 8 snow-classified ($\text{frac_snow} < 0.3$ rule)

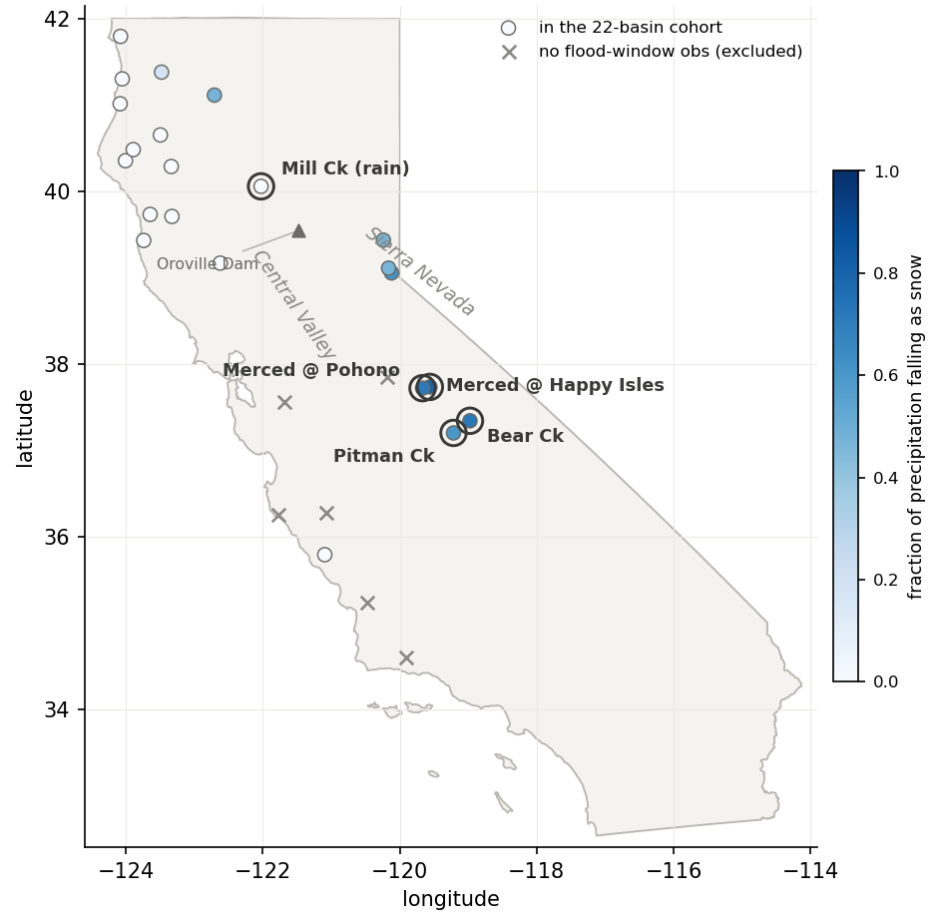


Figure 1: The 28 configured basins, colored by Caravan snow fraction. Circles are the 22-basin evaluation cohort; crosses are the six with no observations in the flood windows. Twenty basins are rain-classified and eight snow-classified under the project's $\text{frac_snow} < 0.3$ rule.

on the reported medians is almost certainly negligible. The split should still have been purged. The overlap is reported in the project’s split-leakage table.

The test period makes a large difference. On 2012–2014, the onset of California’s drought, this architecture scores a median NSE of 0.862. On the two flood years it scores 0.754 (0.836 and 0.679 at a smaller capacity). These are separately trained models with different training years, checkpoints, and forecast realism, so the gap does not isolate the effect of the test window. It is still a useful descriptive comparison, and a reminder that an NSE without its test years says very little.

Finding 1: The baseline that matters is a gauge, not a mean

NSE uses the basin’s mean flow as its built-in baseline, but that is a weak forecast. Two harder baselines were scored on the same dates and matched to each lead. Persistence carries forward the gauge reading from k days earlier. Damped persistence moves that reading toward the seasonal cycle at the basin’s fitted day-to-day autocorrelation. The train-period mean scores -0.08 and the unadjusted seasonal cycle $+0.20$; neither is competitive at short leads.

Table 1: Median NSE by forecast lead, 22-basin cohort and the five focus basins.

Lead (days)	22-basin cohort			5 focus basins		
	LSTM	Persist.	Damped	LSTM	Persist.	Damped
1	0.667	0.712	0.738	0.810	0.944	0.945
2	0.688	0.484	0.566	0.810	0.864	0.869
3	0.665	0.301	0.447	0.809	0.803	0.814
5	0.670	0.037	0.340	0.802	0.702	0.725
7	0.597	-0.122	0.295	0.792	0.610	0.649

At one day, the gauge is better on most basins. The LSTM beats plain persistence on 9 of 22 and damped persistence on 4 of 22. Persistence then deteriorates with lead while the LSTM remains nearly flat. The cohort medians cross at day 2; the LSTM beats plain persistence on 17 basins by day 2, 21 by day 3, and all 22 from day 4 onward. The high-storage focus basins cross later: day 3 against plain persistence and day 4 against damped persistence. Mill Creek, the rain-classified focus basin, behaves differently. The LSTM wins there at every lead, including 0.618 against 0.594 for damped persistence at day 1, because yesterday’s flow contains little information about a flashy rain response.

The comparison was also split by storm size. At lead 2, MAE skill against plain persistence is $+0.06$ in the two largest storm bins, about a third of the pooled value of $+0.17$. By lead 7 it reaches $+0.29$ to $+0.36$. A day-by-day count is less favorable. Above the 95th storm percentile, the LSTM wins 52% of days at lead 2 and 70% at lead 7. In the 80th-to-95th-percentile band, it does not win a majority until lead 4. The early gain in total error comes from avoiding a few large persistence misses, not from winning consistently.

The apparent improvement at longer leads is mostly the failure of persistence. In the largest-storm bin, the LSTM’s error is nearly unchanged from lead 1 to lead 7 (10.9 to 10.8 mm/day, against a bin average of 17.1), while persistence worsens from 10.1 to 15.1. Both forecasts are poor on extreme days, but the LSTM degrades less. Damped persistence also loses its usual

The gauge wins day 1; the model wins the week (crossover day 2-4)

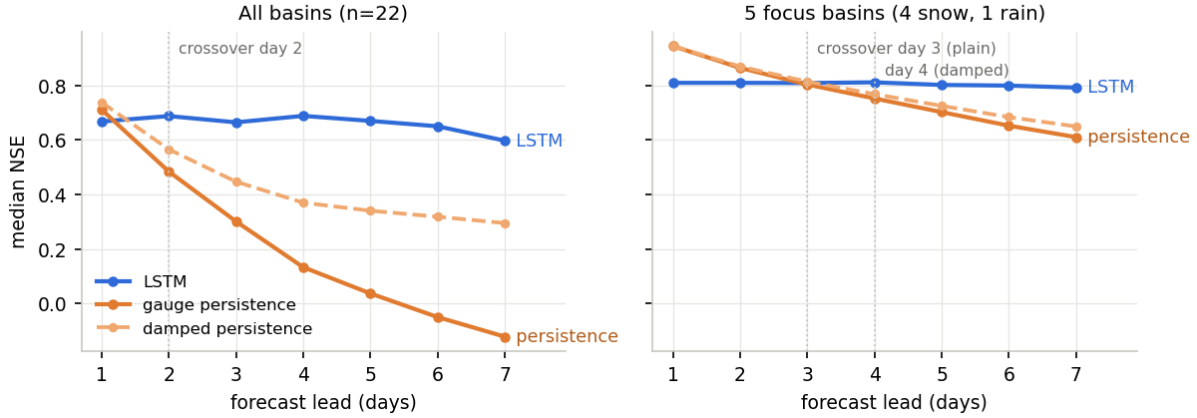


Figure 2: Median NSE against forecast lead for the LSTM, plain gauge persistence, and damped persistence, matched lead for lead. Left: 22-basin cohort. Right: the five focus basins, four snow-classified and one rain.

advantage during storms because it pulls the forecast toward climatology. The stratified table therefore uses whichever persistence baseline is stronger in each bin.

The comparison is not symmetric. Persistence uses observed discharge, which the LSTM never sees and an operator would have. The LSTM, meanwhile, sees forecast weather through the target day. At lead 1 it loses even though it knows about tomorrow’s storm and persistence does not. Most of that loss occurs on quiet days, where MAE skill is -0.46 , compared with -0.08 to -0.15 in the storm bins. Gauge persistence is extremely hard to beat in calm weather and only moderately better during storms. A practical hybrid would give the weather-driven model the latest gauge reading as well.

The seasonal-cycle baseline also changes with the test period: -0.25 on the 2012–2014 drought window and $+0.198$ on the flood years. Even a fixed baseline needs to be read in the context of the years on which it was scored.

Finding 2: The model gets *when*, not *how big*

On the flood years, the deterministic model scores NSE 0.754 and usually places peaks within about a day of their observed timing, but it misses their size. High-flow volume bias—the bias in the top 2% of the flow-duration curve (FHV)—is -19% ; roughly 45% of peaks are missed; and peak magnitude error is about 48%. The timing result should not be overread. Daily means cannot resolve sub-daily timing, and the weather inputs use UTC days while USGS gauges use local days, which can add a one-day offset. “About a day” is the resolution of the data, not a precise timing estimate. Peak counts are also based on only a few events per basin over two water years. For flood operations, the larger problem is magnitude: releases depend on how much water is coming, not simply when the hydrograph rises.

The LSTM is not alone in missing peaks. In a separate January 1997 test, with the entire 1990s removed from training, both the LSTM and NOAA’s process model underestimated the flood peak by tens of percent. The process model came much closer on the rain-driven basin.

LSTM vs gauge persistence by storm size and lead, WY2017+WY2023, cohort n=22
(above zero = LSTM better; valid-date aligned, matched lead)

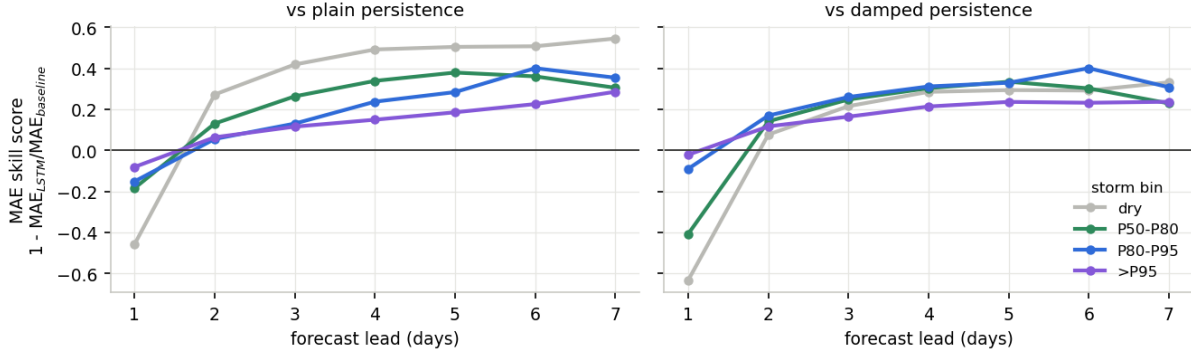


Figure 3: MAE skill score against gauge persistence by storm-size bin and lead, WY2017 and WY2023, 22-basin cohort. Above zero the LSTM has the lower error. The advantage in the top storm bins is smaller than the pooled value at every lead.

The setup favors the LSTM in one respect: 1997 predates the forecast-product archives, so reanalysis—realized future weather—fills its forecast channels. Even with that advantage, the peak remained too low. Peak underestimation at this scale is close to the state of the art for both model families.

Increasing the LSTM’s hidden size from 16 to 128 improved average skill (NSE 0.679 to 0.754) but worsened peak performance. FHV moved from -12% to -19% , and the fraction of missed peaks rose from 0.33 to 0.45. The larger model fit ordinary flows better while showing more regression toward the mean in the tail. Because each capacity was run only once, the pattern is suggestive rather than seed-controlled.

The misses are concentrated in the largest storms. When the 15,907 test basin-days are binned by each basin’s wet-day precipitation climatology, same-day bias stays within $\pm 2\%$ through the 80th percentile. Above the 95th percentile, the model underpredicts flow by 52% and runs low on 84% of days. At three-day lead, matched to conditions on the storm’s arrival day, bias is -30% in the 80th-to-95th-percentile bin and -61% in the top bin; the model runs low on 88% of those days. Error in the top storm bin is almost the same at one-day and seven-day lead. That points to a magnitude problem in the runoff response, not one that a better weather forecast alone would solve.

An event-level plot shows the same pattern but remains exploratory. Across 281 observed peaks, the median error falls from -13% for events below a quarter of the basin’s training-record maximum, to -26% and -33% in the middle bins, and -72% in the top bin. The top bin contains only three events. Events also cluster by basin and storm sequence, and the observed peak appears on both axes, which can create a negative slope by construction. The storm-percentile analysis above avoids those problems and is the firmer result.

Held-out WY2017 (Oroville AR season): timing is right, magnitude is not

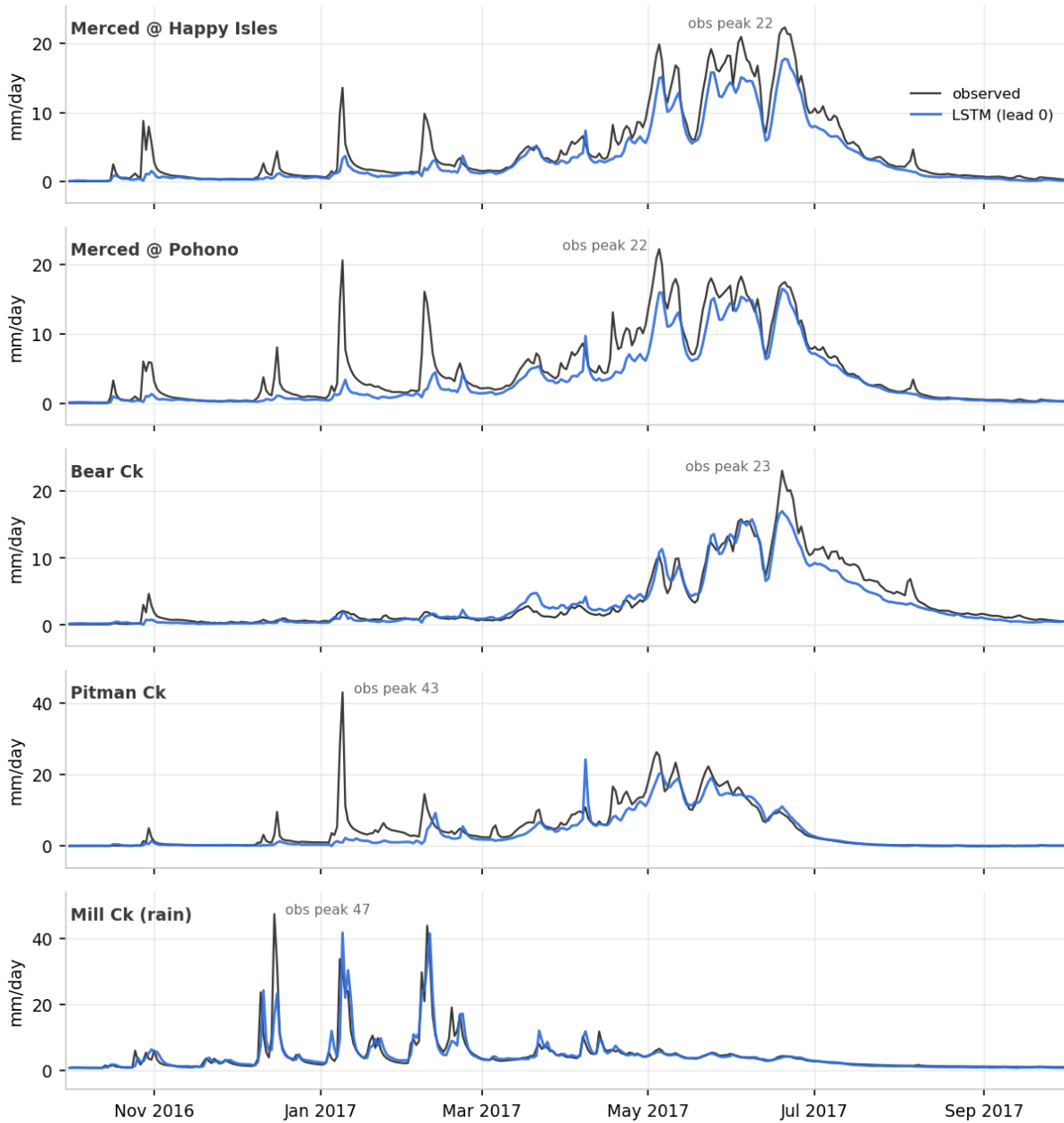


Figure 4: Observed and simulated daily flow for the focus basins in the held-out WY2017 window. The LSTM follows the broad timing but underestimates several peaks.

Skill by storm-size bin (per-basin wet-day precipitation percentile), WY2017 + WY2023, cohort n=22
(obs, precipitation and storm class all taken at valid date = issue date + lead)

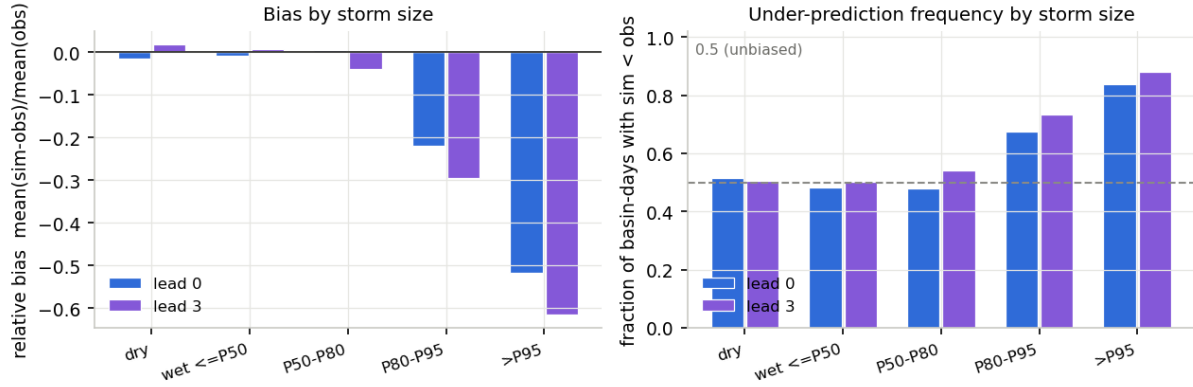


Figure 5: Bias and under-prediction frequency by storm-size bin, at same-day and three-day lead. Each lead is scored at its own valid date, so the three-day rows are conditioned on the storm's arrival day.

Peak error vs. event out-of-distribution-ness, 281 events, 22 basins, WY2017+WY2023 (min peak distance 5d)

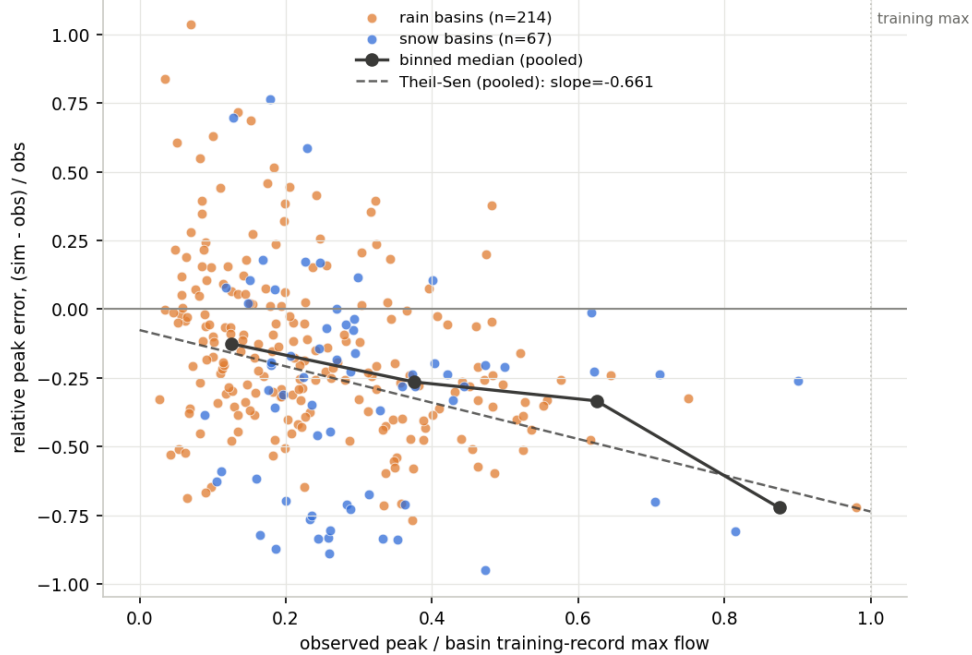


Figure 6: Relative peak error against event magnitude, normalized by each basin's training-record maximum flow. Exploratory: the top bin holds three events and the observed peak appears in both axes.

Finding 3: A held-out basin failed, and the experiment cannot say why

Most basins are ungauged, so the model was retrained while leaving each focus basin out in turn. Bear Creek, a snow-classified basin, drops to NSE 0.646 with high-flow bias of -29% . Pitman Creek, also snow-classified, drops to 0.375 with -62% bias. Mill Creek, the rain-classified basin, falls to -0.740 and its bias flips to $+80\%$. The snow basins lose skill; Mill Creek becomes worse than predicting its observed mean and overpredicts the flows that the full model underpredicts.

Rain-driven Mill Ck, WY2023: held out of training, skill inverts (LOBO experiment)

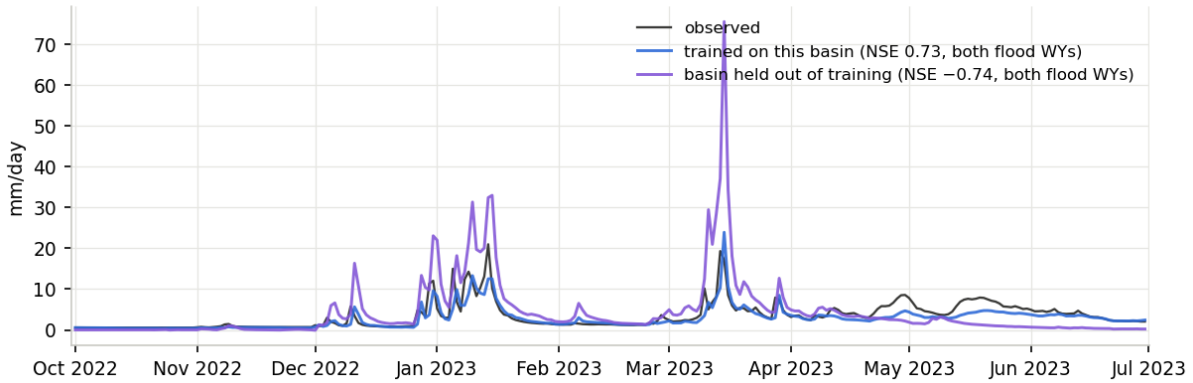


Figure 7: Mill Creek in WY2023, simulated by the model trained on all 28 basins and by the model with Mill Creek held out. NSE values in the legend are computed over both flood water years.

This experiment cannot explain the failure. An earlier draft described it as transfer from snowmelt to rainfall, but the training set does not support that account. After Mill Creek is removed, 19 of the remaining 27 basins are rain-classified, and eighteen share Mill Creek’s snow fraction of zero. With only one failed basin, the experiment cannot distinguish among differences in basin attributes, catchment scale, precipitation-to-flow timing, record extension, spatial extrapolation, or training variance. Published regionalization results use hundreds of hydrologically diverse basins (Kratzert et al., 2019; Nearing et al., 2024). Identifying a mechanism here would require leave-cluster-out tests across several basin groups.

Even without a diagnosis, the result matters operationally. The regional model failed completely on an unseen basin, and its output gave no warning. Strong performance on the training cohort did not protect Mill Creek.

Two further basins, the nested Merced pair, scored 0.857 and 0.790 in the same experiment. Each has its nested partner in the training set, which leaks the hydrograph, so they are excluded from the ungauged claim.

Finding 4: Sharper but overconfident

The distributional head predicts a probability distribution for each day’s flow instead of a single value. It is compared with the deterministic head using the continuous ranked probability score

(CRPS) (Gneiting and Raftery, 2007). For a single-value forecast, CRPS is exactly absolute error. The same proper score, used here in its fair finite-ensemble form (Ferro, 2014), can therefore test whether the predicted spread adds useful information.

Table 2: Median CRPS (mm/day) over the 22-basin cohort. Lower is better.

Lead (days)	Distributional	Deterministic	Improvement
0 (same-day)	0.916	1.231	26%
3	1.081	1.452	26%
7	1.185	1.550	24%

It does. The distributional head reduces CRPS by 22–26% at every lead. It also performs slightly better on several point metrics: NSE rises from 0.754 to 0.784, with fewer missed peaks and smaller peak error, although KGE declines.

The distributional head: sharper and uncalibrated

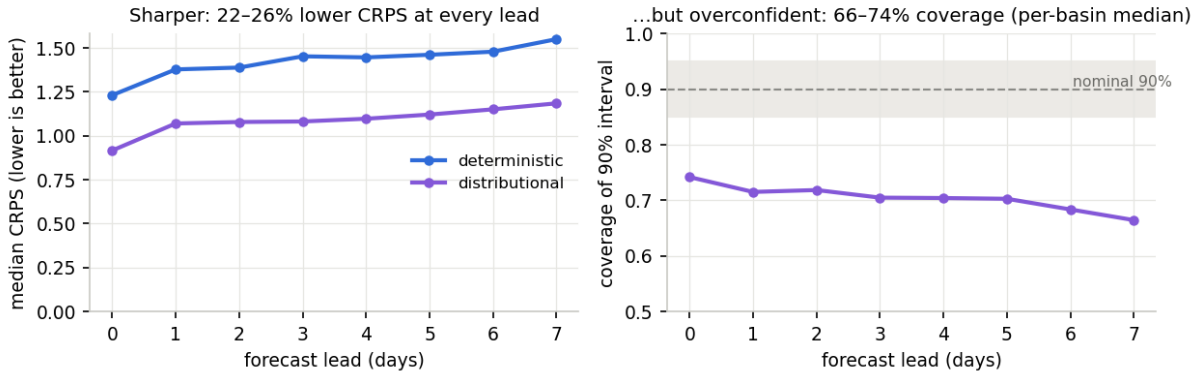


Figure 8: Left: CRPS against lead for both heads. Right: empirical coverage of the nominal 90% predictive interval against lead, as a median over per-basin coverages.

The intervals are too narrow. Nominal 90% intervals contain only 66–72% of observations, with little change from lead 0 to lead 7. For an operator, this kind of overconfidence is more dangerous than an interval that is too wide. Across 15,907 same-day basin-days, the rank histogram is roughly symmetric and U-shaped: 22% of observations lie above the ensemble’s 90th-percentile rank and 20% below its 10th. That is the signature of underdispersion. Conditioning on flow separates the tails. At high flows, 23% of observations escape above the interval and 3% below it; at low flows, 23% escape below and 9% above. The predictive distribution is pulled toward middle flows at both ends. A separate heavy-tail problem, in which rare samples inflate the ensemble standard deviation to roughly three times the RMSE, explains the spread/skill ratio and NaN training losses, but not the poor coverage.

An affine calibration was tested that shifts the median and rescales the spread within bands of predicted flow. It is fit on one flood water year and applied to the other, so every reported value is out of sample. Coverage rises from 0.66–0.72 to 0.88–0.91 at every lead, with almost no change in sharpness. The flow-conditional version leaves same-day CRPS unchanged (0.901 raw, 0.900 corrected) and improves it by about 1% at seven days (1.209 to 1.196). A single global

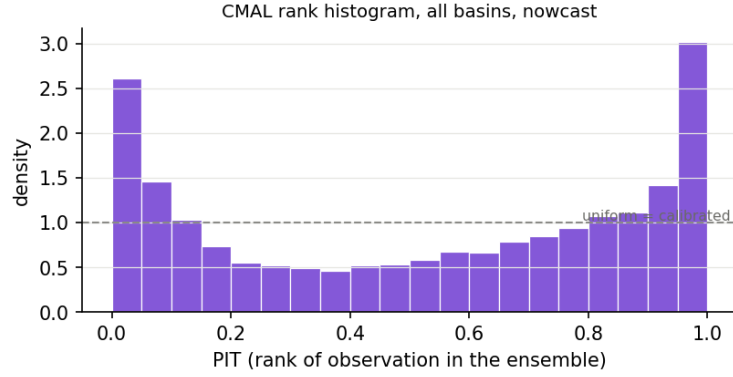


Figure 9: Rank histogram of the observation within the predictive ensemble, all basins, lead 0. The dashed line is the uniform density of a calibrated ensemble.

correction costs 0.00–0.04 mm/day of CRPS at every lead. In the highest-flow third of days, raw coverage is 0.70 and 26% of observations lie above the interval. The flow-conditional map reaches 0.85 coverage, cuts the upper miss rate to 12%, and has the best CRPS of the three variants (2.78, compared with 2.89 global and 2.85 raw). The global map gives higher coverage there, at 0.89. The cross-winter result matters operationally: a correction fit on one atmospheric-river winter transfers to another.

Coverage and fair CRPS by lead, flood test years: raw ensemble vs affine recalibration, each window scored with parameters fit on the other (scored at valid date = issue date + lead)

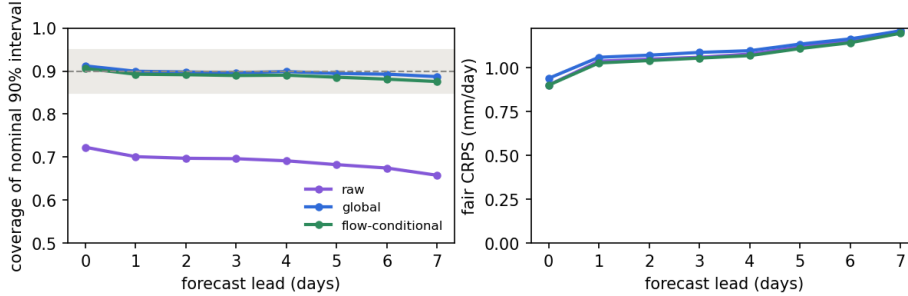


Figure 10: Coverage and fair CRPS against lead on the flood test years for the raw ensemble and two affine corrections, each water year scored with parameters fit on the other.

An earlier version reported 0.46 coverage at day 7 and an 11% long-lead improvement in CRPS after calibration. Both were artifacts of a scoring bug: multi-lead forecasts had been paired with observations from the issue date instead of the target date. The results were corrected on 2026-08-19 after an adversarial review found the error; the last section explains what happened. The corrected result is simpler. Raw coverage is low by a similar amount at every lead, and affine calibration fixes it without a meaningful loss of sharpness.

Finding 5: The benchmark win depends on the benchmark

The comparison model is NOAA’s National Water Model (NWM), the continental process simulation behind federal streamflow guidance. One qualification matters before the scores: the comparison uses NWM retrospectives, which rerun past decades with observed weather and no gauge assimilation, and compare them with the LSTM’s same-day hindcast. This is a comparison between simulations, not between operational forecasts. The operational NWM assimilates gauge data and would be a different benchmark.

The comparison that motivated this project used 2012–2014. There, the LSTM beat the v2.1 retrospective on every focus basin, with median NSE of 0.83 against 0.53. The comparison was rerun on the flood years with both v2.1 and the current v3.0:

Table 3: LSTM against two NWM retrospective versions on the flood windows, five focus basins.

Window	LSTM	NWM v2.1	NWM v3.0
WY2017, median NSE	0.772	0.628	0.848
WY2017, high-flow bias	−20%	−2%	−5%
WY2023 (Oct–Jan), median NSE	0.443	—	0.162

On WY2017, NWM v3.0 beats the LSTM on median skill and on three of the five basins, with much smaller high-flow bias and peak error. Much of the LSTM’s original margin was therefore specific to NWM v2.1. The v3.0 retrospective ends on 2023-02-01, so the WY2023 comparison covers the December–January storms but not the storms in March. On that partial window, the LSTM has higher median skill, while both systems underestimate high flows by 53–55%.

The benchmark win depends on the benchmark: NWM v3.0 leads WY2017 (3 of 5 basins); the LSTM leads WY2023 and Bear Ck

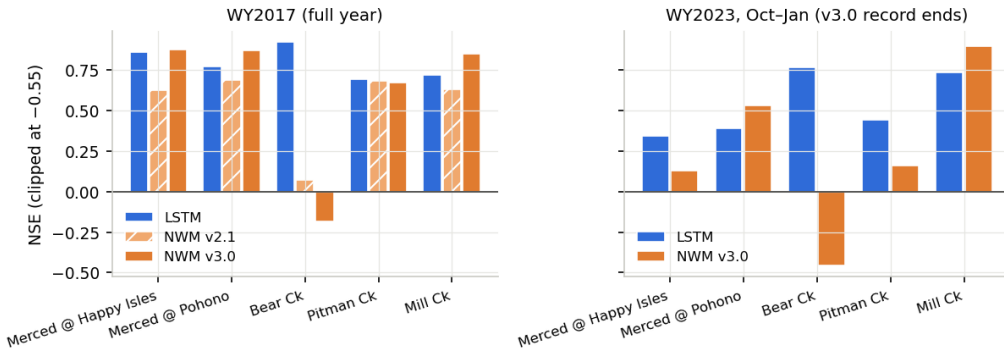


Figure 11: Per-basin NSE for the LSTM and the two NWM retrospective versions on the flood windows.

Two basin-level patterns recur in every window tested. The LSTM wins Bear Creek, a high-elevation, high-storage snow basin, by a wide margin in both flood years (0.92 and 0.77, compared with at best 0.07 for the NWM). The NWM does much better on rain-driven flood peaks. At Mill Creek in WY2023 it scores 0.90 against 0.74 for the LSTM, with a similar pattern in 1997.

These results do not support a general claim that machine learning beats the process model.

The ranking depends on the NWM version, the test period, and the basin. What the comparison does show is that the models fail in different places.

A note on the winter ahead

Written in August 2026; unlike the results above, this section will date quickly.

NOAA’s Climate Prediction Center reports El Niño conditions, and its August outlook gives a very strong event 90–95% odds through early winter (NOAA Climate Prediction Center, 2026). Strong El Niño winters tend to bring a southward-displaced subtropical jet and warmer atmospheric rivers with high snow levels, putting more rain on basins that usually receive snow (Gershunov et al., 2025). Those conditions line up with several weaknesses found here. The model underpredicts flow by 52% above the 95th storm percentile at same-day lead and by 61% at three-day lead. Warm storms also move snow basins away from their usual behavior, while the held-out-basin test shows that failure outside the training envelope may not announce itself. At high flows, raw predictive intervals miss above the interval seven times as often as below; they should be calibrated before use. The most defensible part is the two-to-seven-day window, where the model generally beats the gauge, although only by a small margin on the largest storm days. That is also the FIRO decision window, and a weather-driven model can make direct use of better atmospheric-river forecasts from reconnaissance flights.

What this adds up to

Table 4: Verdicts by use case.

Use case	Verdict
2–7 day volume forecast, gauged basin, conditions resembling training	Yes. Beats both persistence baselines on the cohort median from day 2 and on all 22 basins from day 4, across two challenge flood years. Within the top storm bins the margin thins to about a third of the pooled value but holds
1-day forecast	Mostly no. Beats plain persistence on 9 of 22 basins and damped on 4 of 22; the gauge is better, and free. Flashy rain basins are the exception
Flood peak magnitude	Not yet. The process model shares much of the same failure
Ungauged basin	Not demonstrated. One held-out basin inverted (NSE -0.74), and this experiment does not establish why
Calibrated uncertainty	Raw, no (66–72% at a claimed 90%). After cross-fitted calibration, yes (88–91% at every lead, CRPS-neutral), tested across winters
Replacing the process model	No. The v3.0 retrospective wins WY2017 on 3 of 5 basins; the models are complementary by regime

The main usable result is a measured window of forecast value: for gauged basins under

conditions similar to the training data, the LSTM improves on the operator’s gauge baseline at two-to-seven-day leads. It does not establish that machine learning should replace the process model.

What it took to trust these numbers

One scoring protocol is used throughout. Simulations come from each run’s inference output, while observations come directly from the gauge records rather than the copy stored with a model run. Metrics use the framework’s implementations, checkpoints are selected on validation skill, and medians use the stated 22-basin cohort. Deterministic inference reproduced bit for bit across repeated runs. For the distributional model, bootstrap uncertainty in the sample-median metrics is below 0.001 NSE, so ensemble sampling does not explain the reported differences. Training variance remains uncontrolled because each configuration was run once; small margins should not be read to the third decimal.

The protocol became strict because closure tests found several problems. One choice of test period raised skill by 0.11 NSE. An evaluation path silently dropped more than a quarter of the scoreable basins from its medians. An upstream bug wrote corrupted observations into the distributional model’s output files; it was found because two metrics reported for the same simulation were mutually inconsistent. Comparing the extended gauge records with their overlap in the original data exposed a catchment-area discrepancy and five gauges whose records disagree outright. Those gauges were excluded instead of rescaled.

The protocol still missed one important error. In August 2026, an adversarial review found that two multi-lead analyses paired forecasts with observations from the issue date rather than the target date. A seven-day forecast was therefore scored against flow from seven days too early. Findings 2 and 4 now use the corrected recalibration and lead-3 storm results. The earlier version exaggerated the decline in interval coverage with lead, created an apparent CRPS gain from recalibration, and understated storm-tail bias. There was a clue that should have raised concern: it showed lower CRPS at one-day lead than at same-day, even though both forecasts used the same information.

A closure test now checks every lead against the source gauge record on the target date; the values agree to one float32 bit. The 28 overlapping boundary dates in the train/test split are also tabulated. The same review corrected the project’s earlier Sierra snowmelt framing to the 20-rain, 8-snow basin census used here and removed a synthetic-storm experiment whose causal interpretation did not hold up. That experiment remains in the project log as an exploratory diagnostic.

Known limitations, failure modes, and the traps encountered along the way are documented in `docs/METHODS.md`.

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